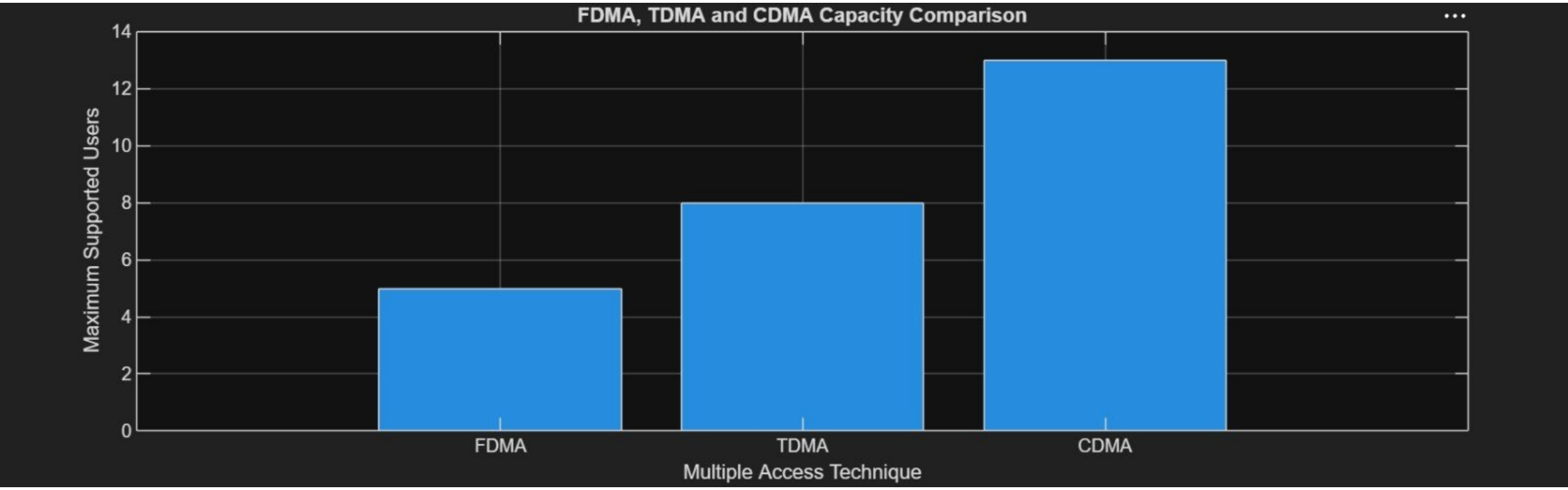
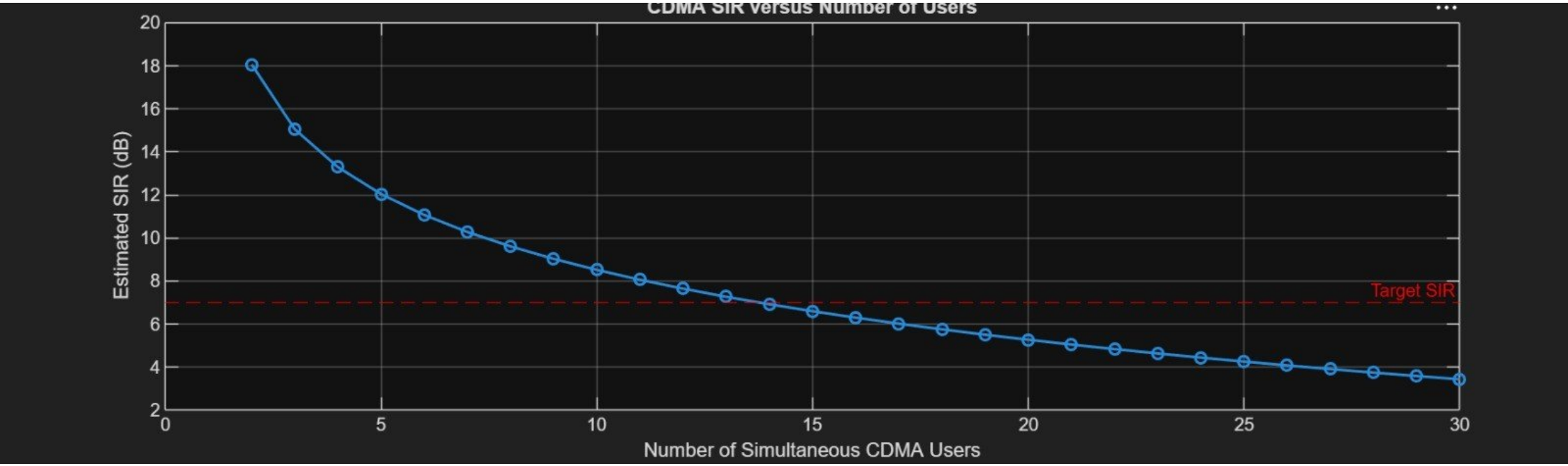
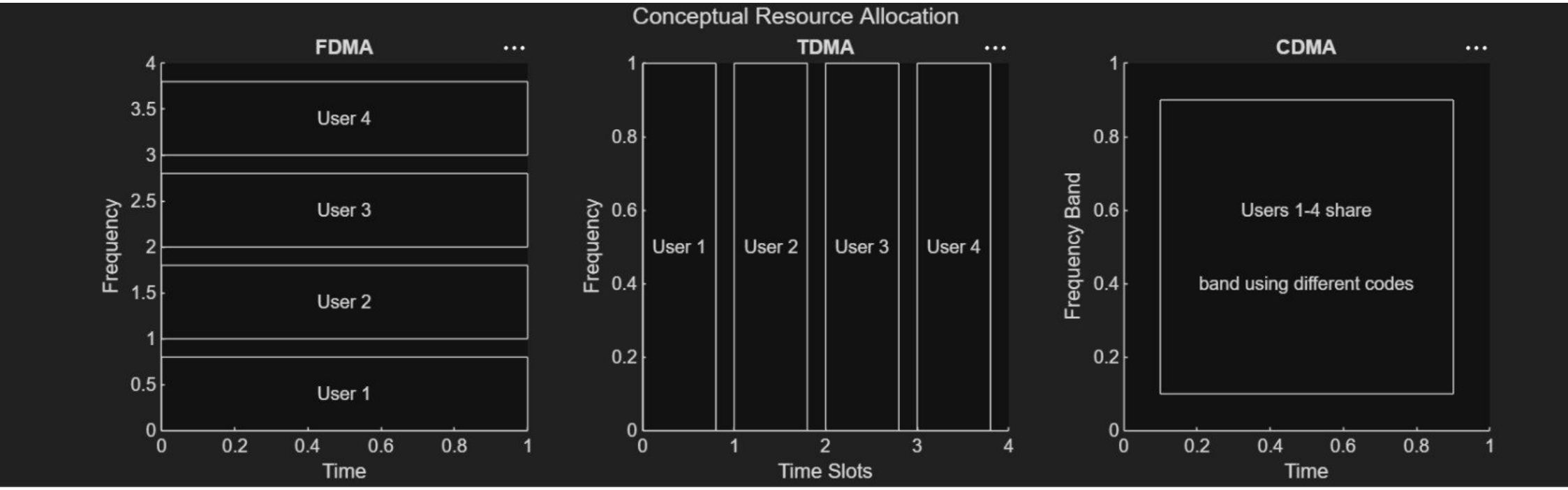
```
clear;
clc;
close all;
% System Parameters
totalBandwidthHz = 1e6;
fdmaChannelHz = 200e3;
tdmaSlots = 8;
processingGain = 64;
targetSIRdB = 7;
% Calculate User Capacity
fdmaUsers = floor(totalBandwidthHz/fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB/10);
cdmaUsers = floor(1 + processingGain/targetSIR);
% Display Results
fprintf('Maximum Supported Users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users\n', cdmaUsers);
% Graph 1: Capacity Comparison
figure;
bar([fdmaUsers tdmaUsers cdmaUsers]);
set(gca,'XTickLabel',{'FDMA','TDMA','CDMA'});
xlabel('Multiple Access Technique');
ylabel('Maximum Supported Users');
title('FDMA, TDMA and CDMA Capacity Comparison');
grid on;
% Graph 2: CDMA SIR
K = 2:30;
sirLinear = processingGain./(K-1);
```

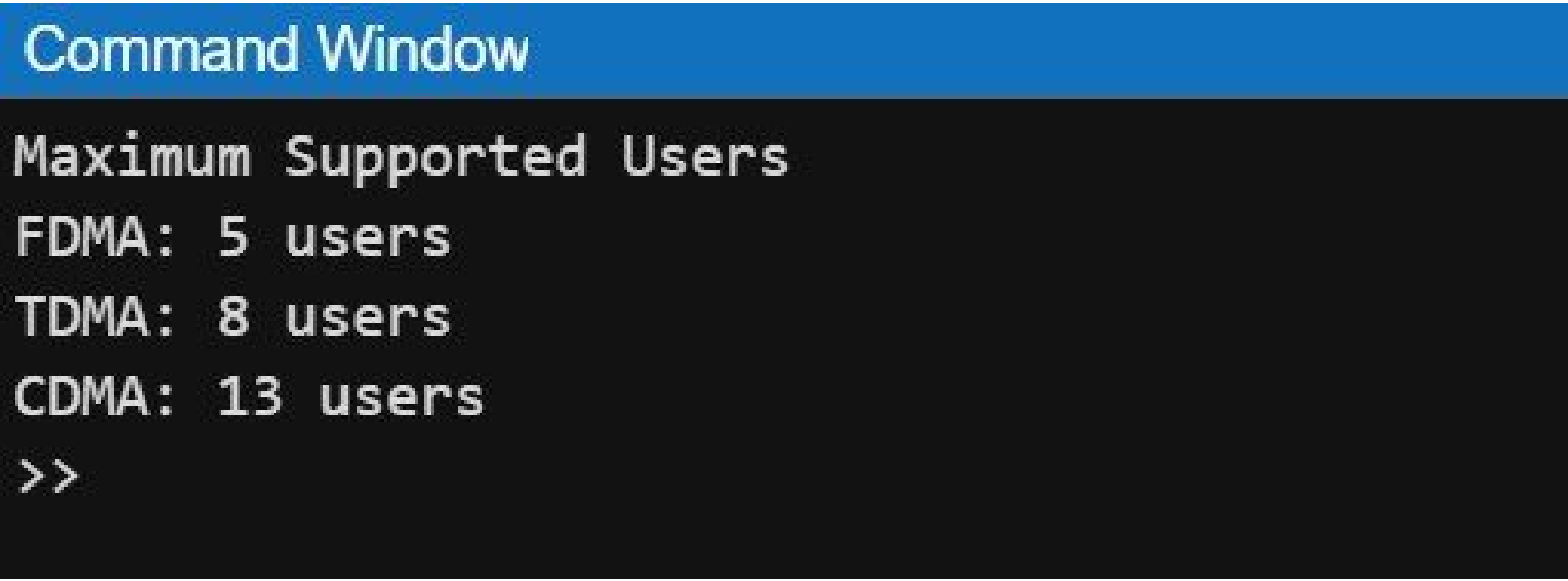
```
sirLinear = processingGain./(K-1);
sirDb = 10*log10(sirLinear);
figure;
plot(K,sirDb,'o-','LineWidth',1.5);
hold on;
yline(targetSIRdB,'--r','Target SIR');
xlabel('Number of Simultaneous CDMA Users');
ylabel('Estimated SIR (dB)');
title('CDMA SIR versus Number of Users');
grid on;
hold off;
% Graph 3: Resource Allocation
figure;
% FDMA
subplot(1,3,1);
hold on;
for u = 1:4
    rectangle('Position',[0 u-1 1 0.8]);
    text(0.5,u-0.6,sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 1]);
ylim([0 4]);
xlabel('Time');
ylabel('Frequency');
title('FDMA');
hold off;
% TDMA
subplot(1,3,2);
hold on;
```

```
title('FDMA');
hold off;
% TDMA
subplot(1,3,2);
hold on;
for u = 1:4
    rectangle('Position',[u-1 0 0.8 1]);
    text(u-0.6,0.5,sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 4]);
ylim([0 1]);
xlabel('Time Slots');
ylabel('Frequency');
title('TDMA');
hold off;
% CDMA
subplot(1,3,3);
rectangle('Position',[0.1 0.1 0.8 0.8]);
text(0.5,0.6,'Users 1-4 share', ...
    'HorizontalAlignment','center');
text(0.5,0.4,'band using different codes', ...
    'HorizontalAlignment','center');
xlim([0 1]);
ylim([0 1]);
xlabel('Time');
ylabel('Frequency Band');
title('CDMA');
sgtitle('Conceptual Resource Allocation');
```


FDMA, TDMA and CDMA Capacity Comparison
Maximum Supported Users
0
2
4
6
8
10
12
14
FDMA
TDMA
CDMA
Multiple Access Technique


CDMA SIR Versus Number of Users
Estimated SIR (dB)
Number of Simultaneous CDMA Users
Target SIR
20
18
16
14
12
10
8
6
4
2
0
5
10
15
20
25
30


Conceptual Resource Allocation
FDMA
User 4
User 3
User 2
User 1
Frequency
Time
TDMA
User 1
User 2
User 3
User 4
Frequency
Time Slots
CDMA
Users 1-4 share
band using different codes
Frequency Band
Time


Command Window
Maximum Supported Users
FDMA: 5 users
TDMA: 8 users
CDMA: 13 users
>>